

Missing Value Identification

Task 12 | Data Analytics Track | Veda Technology Internship

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1. Objective

The goal of this task was to find missing values in a dataset and summarize where they occur. This is a basic data inspection step done before any cleaning or analysis.

2. Dataset and Tools

I used the Titanic dataset, which has **891 rows** and **12 columns**. I chose it over Iris because Iris has no missing values, while Titanic has real missing data to practice on.

- Google Colab (notebook environment)
- Python and Pandas (null checks and summary table)
- Matplotlib (bar chart)

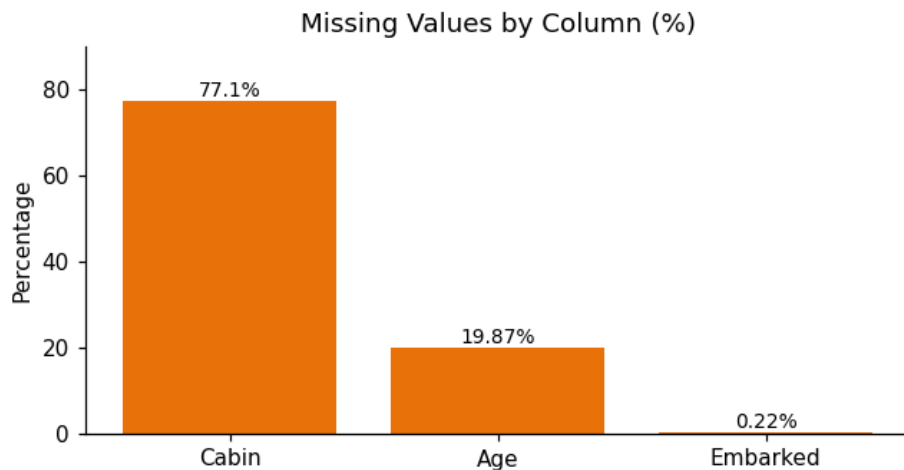
3. Method

- Loaded the CSV file with `pd.read_csv()` and checked the shape and first rows.
- Used `df.info()` to see data types and non-null counts.
- Counted nulls per column with `df.isnull().sum()` and calculated the percentage of each.
- Kept only the columns with missing values and sorted them from highest to lowest.
- Plotted the percentages as a bar chart and saved the summary to a CSV file.

4. Missing Value Summary

Column	Missing Count	Missing %
Cabin	687	77.1%
Age	177	19.87%
Embarked	2	0.22%

The other 9 columns have no missing values.



5. Findings

- **Cabin** has the most missing values (687 of 891, about 77%). Most passengers have no cabin number recorded, so this column is not very useful as it is.
- **Age** has 177 missing values (about 20%). This is a big gap, but age is important, so it can be filled later, for example with the median.
- **Embarked** has only 2 missing values (about 0.22%), so its effect on the data is very small.

6. Conclusion

Only 3 out of 12 columns have missing values. No values were removed or filled in this task, because the goal was only to identify them. Deleting rows without a proper reason could lose useful data, so the handling method will be decided in the next steps.